

Monetary Economics and Policy

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Chapter 11: Shortage of "Safe
Assets"



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What We Will Learn

- ▶ What economists mean by “**safe assets**”: debt securities that (i) retain value in the **unit of account** and (ii) provide **liquidity services** in transactions and collateral.
- ▶ Why a **shortage of safe assets** shows up as a **liquidity premium** and persistently low safe interest rates.
- ▶ How **private intermediaries** can create **safe liabilities** in frictionless settings, and why this does **not** necessarily undermine **price-level control**.
- ▶ How a realistic friction, **costly monitoring** of safe investments, generates **pseudo-safe** (risky but liquid) liabilities and a **liquidity crunch** in bad states.
- ▶ How the **policy rate** can still stabilize prices, yet cannot by itself prevent a **collapse in real liquidity** when pseudo-safe assets default.
- ▶ How policy can prevent a liquidity crunch: by expanding **public liquidity** and/or backstopping private money-like claims through **guarantees** and regulation.

Outline

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways
- 10 Bibliography

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Safe Assets: Definition and Liquidity Premium

- ▶ **Safe assets** are financial securities that:
 - ▶ retain value in terms of the currency over time,
 - ▶ provide **liquidity services**: used for transactions and as collateral.
- ▶ These services generate a **liquidity premium**:
 - ▶ safe assets carry **lower interest rates** than otherwise comparable securities,
 - ▶ the issuer finances itself on more favorable terms.

Public vs. Private Production of Safe Assets

- ▶ **Public production:** the central bank is the natural provider.
 - ▶ In a fiat system, central bank liabilities define the unit of account and are inherently safe.
 - ▶ Yet their supply is often **limited** relative to liquidity needs:
 - ▶ domestic demand for transaction and collateral services,
 - ▶ especially global demand when the currency is a reserve asset (e.g. the dollar).
- ▶ **Private production:** intermediaries can bridge the gap by issuing money-like liabilities backed by assets and equity buffers.

Another Candidate: The Sovereign

- ▶ **Sovereign debt** can be a safe asset, but not for every sovereign:
 - ▶ safety is an **institutional achievement**, often preceded by repeated defaults,
 - ▶ credible repayment requires the capacity to **levy taxes** (Hamilton 1947; Gorton 2017).
- ▶ **Consolidated view** (Treasury + central bank):
 - ▶ total government liabilities must be backed by taxes and/or by assets held on the public balance sheet,
 - ▶ the riskiness of private assets held by the public sector affects the feasible tax burden.

2007–2008: Pseudo-Safe Assets and Liquidity Collapse

- ▶ **Pre-crisis:** limited supply of public safe assets encouraged private creation of money-like claims (shadow banking).
- ▶ **Crisis lesson:** many private claims were only **pseudo-safe**:
 - ▶ liquid in good times but insufficiently backed,
 - ▶ in the crisis they lost both value and **liquidity services**.
- ▶ **Macro repercussions:**
 - ▶ financial “disequilibria” mirror goods-market disequilibria,
 - ▶ a **shortage of safe assets** corresponds to an **excess supply of goods**, generating falling prices and/or recession.

A Friction in Private Safe-Asset Production

- ▶ **Monitoring cost:** it is costly to identify and monitor safe investments that back private liquidity.
- ▶ **Financing monitoring:** intermediaries rely on the **liquidity premium**:
 - ▶ the premium depends on the supply of **government liquidity**,
 - ▶ high public liquidity $\Rightarrow$ low premium $\Rightarrow$ **crowd-out** of safe private production.
- ▶ **Risky debt persists:**
 - ▶ risky debt is cheaper to produce (no monitoring) and can be issued abundantly,
 - ▶ when used for liquidity, it becomes **pseudo-safe**: liquid in favorable states, but prone to a liquidity collapse in adverse states.

Macroeconomic Impact of a Liquidity Crunch

- ▶ When pseudo-safe assets lose backing, they default and cease to provide liquidity services:
 - ▶ a **safe-asset shortage** emerges,
 - ▶ liquidity premia and borrowing costs rise.
- ▶ A collapse in money-like claims can translate into a contraction in real activity (in the spirit of Friedman and Schwartz (1963)).
- ▶ **Policy objective:** keep nominal spending (or broad monetary aggregates) on its pre-crisis path by offsetting the collapse in private liquidity with public intervention.
- ▶ Targeting nominal spending avoids having to identify every evolving form of quasi-money.

Post-2008 Policy Responses

- ▶ **Public provision:** central bank asset purchases (QE) expanded reserves to satisfy demand for safe/liquid claims.
- ▶ **Government guarantees:** deposit insurance expansions and programs (e.g. FDIC guarantees) that “harden” private liabilities.
- ▶ **Regulation:** tighter capital/liquidity requirements aimed at making intermediaries’ debt safer and reducing crisis likelihood.

- 1 Introduction
- 2 Outline of the results**
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Outline of the Results

- ▶ **Frictionless benchmarks**
 - ▶ **Public supply:** with unconstrained lump-sum taxes, the government can satiate liquidity and pin down the price level.
 - ▶ **Private supply:** with no monitoring costs ($\iota = 0$), competitive intermediaries supply enough safe private debt to satiate liquidity in all states.
- ▶ **Monitoring costs and pseudo-safety**
 - ▶ **Shortage of private safe assets:** when monitoring is costly ($\iota > 0$), safe private issuance is limited and intermediaries issue risky debt.
 - ▶ **Pseudo-safe assets:** risky debt is liquid in good states but defaults in bad states, generating an endogenous liquidity crunch.
- ▶ **Intervention and regulation**
 - ▶ **Asset purchases / guarantees:** can eliminate crises and economize on taxes in good states, but crowd out truly safe private debt.
 - ▶ **Regulation:** forcing intermediaries to hold only riskless assets removes crises but is welfare-inferior because it wastes monitoring resources.

- 1 Introduction
- 2 Outline of the results
- 3 Model**
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Model Overview: Structure and Risk

- ▶ **Timing:** two periods, t and $t + 1$.
- ▶ **Agents:** households, financial intermediaries, and the government.
- ▶ **Aggregate risk at $t + 1$:** two states of nature

$$s \in \{h, l\}, \quad \Pr(h) = 1 - \kappa, \quad \Pr(l) = \kappa, \quad 0 < \kappa < 1.$$

- ▶ **Interpretation:** the low state l captures a systemic event: risky investments fail, intermediaries default on risky liabilities, and the economy experiences a **liquidity crunch**.

Financial Intermediaries: Technologies

At time t , intermediaries choose between two distinct investment opportunities:

- ▶ **Safe Technology:**

- ▶ A riskless investment that maintains its value in **both states** at $t + 1$.
- ▶ Backs **safe private debt** which is never defaulted on.

- ▶ **Risky Technology:**

- ▶ Productive in the high state (h), but **loses all value** in the low state (l).
- ▶ Backs **risky private debt** which undergoes full default in state l .

Households: Assets and State-Contingent Payoffs

Households can invest at time t in three securities (zero-coupon, face value 1) with the following payoffs:

- ▶ **Government debt** B_t : Always repaid (safe).
- ▶ **Safe private debt** S_t : Backed by safe projects; never defaults.
- ▶ **Risky private debt** D_t : Subject to state-contingent default:

$$\text{payoff}(D_t) = \begin{cases} 1 & \text{in state } h, \\ 0 & \text{in state } l. \end{cases}$$

Endogenous Liquidity Crisis

Realized Liquidity Services at $t + 1$

- ▶ Liquidity benefits enter the utility function via the function $V(\cdot)$.
- ▶ **High State (h):** All assets are solvent. Realized liquidity is:

$$(B_t + S_t + D_t)/P_h$$

- ▶ **Low State (l):** Risky debt defaults and ceases to provide services.

$$(B_t + S_t)/P_l$$

Key Assumptions on Timing

- ▶ Liquidity benefits are provided in period $t + 1$ (rather than t as in Chapter 4).
- ▶ Therefore, this assumption is key to modelling a liquidity crisis that occurs when the economy switches to the low state.

Households: Preferences

Expected Intertemporal Utility

$$C_t + \beta(1 - \kappa) \left[V \left(\frac{B_t + S_t + D_t}{P_h} \right) + C_h \right] + \beta\kappa \left[V \left(\frac{B_t + S_t}{P_l} \right) + C_l \right]. \quad (11.1)$$

Interpretation

- ▶ C_t, C_h, C_l : Consumption at t , and in states h, l at $t + 1$, providing linear utility.
- ▶ $\beta \in (0, 1)$: Discount factor; κ : Probability of the low state.
- ▶ $V(\cdot)$: Concave utility from real liquid services with satiation point $\bar{q} > 0$.
- ▶ B_t : Government bonds; always risk-free.
- ▶ S_t : Private Safe Assets; backed by safe technologies (no default).
- ▶ D_t : Private Risky Assets; provide liquidity in state h , but default in state l .

Households: Budget Constraint at time t

Budget Constraint at time t

$$Q_t^B B_t + Q_t^S S_t + Q_t^D D_t + P_t C_t \leq P_t Y_t + B_{t-1}. \quad (11.2)$$

Interpretation

- ▶ Q_t^B, Q_t^S, Q_t^D : Prices of public, safe private, and risky private debt.
- ▶ Y_t : Endowment of goods at time t .
- ▶ B_{t-1} : Initial holdings of government bonds.

Note

- ▶ Prices Q_t represent the discount; lower prices imply higher yields.
- ▶ Consistent with Chapter 1, we work with prices rather than gross interest rates for tractability.

Households: State-Contingent Constraints at $t + 1$

Budget Constraints at time $t + 1$

$$P_h C_h \leq P_h Y_h + B_t + S_t + D_t + \Psi_h^A - T_h, \quad (11.3)$$

$$P_l C_l \leq P_l Y_l + B_t + S_t + \Psi_l^A - T_l. \quad (11.4)$$

Interpretation

- ▶ Y_h, Y_l : State-contingent endowments at $t + 1$.
- ▶ T_h, T_l : State-contingent lump-sum taxes.
- ▶ Ψ_h^A, Ψ_l^A : Aggregate profits from financial intermediaries.

Note

- ▶ Importantly, complete default on D_t is an **equilibrium outcome**, not a mere assumption.

Consumption and Portfolio Choice

Optimization Problem

- ▶ Households choose $\{C_t, C_h, C_l, B_t, S_t, D_t\}$ to maximize intertemporal utility (11.1) subject to budget constraints (11.2), (11.3) and (11.4).

The Linear Utility Simplification

- ▶ Utility is assumed to be **linear** in consumption C .
- ▶ **Implication:** As in Chapter 4, this simplifying assumption always anchors the real interest rate to the discount factor:

$$1 + r = \frac{1}{\beta}.$$

Households: First-Order Conditions

State-Contingent Liquidity Premia

$$\varrho_h \equiv V_q \left(\frac{B_t + S_t + D_t}{P_h} \right), \quad \varrho_l \equiv V_q \left(\frac{B_t + S_t}{P_l} \right).$$

FOC for Government Bonds:

$$Q_t^B = \beta \left[(1 - \kappa) \frac{P_t}{P_h} (1 + \varrho_h) + \kappa \frac{P_t}{P_l} (1 + \varrho_l) \right]. \quad (11.5)$$

FOCs for Private Securities:

$$Q_t^S \geq \beta \left[(1 - \kappa) \frac{P_t}{P_h} (1 + \varrho_h) + \kappa \frac{P_t}{P_l} (1 + \varrho_l) \right], \quad (11.6)$$

$$Q_t^D \geq \beta \left[(1 - \kappa) \frac{P_t}{P_h} (1 + \varrho_h) \right]. \quad (11.7)$$

Implications of Household FOCs

Equilibrium Interpretation

- ▶ (11.5) holds with equality: since B_t is in positive supply, it is always utilized for liquidity.
- ▶ (11.6) and/or (11.7) hold with equality when these securities are issued and used for liquidity.

Pricing Implications

- ▶ D_t provides services only in state h (the non-default state).
- ▶ Comparing (11.6) and (11.7) implies $Q_t^S \geq Q_t^D$ (safe assets command a higher price/lower yield).
- ▶ Liquidity services also lower the borrowing costs for issuers.

Intermediaries: Technologies and Monitoring Cost

Capital Types:

- ▶ **Safe capital** K_t^S : requires extra resource cost ι per unit (*monitoring cost*).
- ▶ **Risky capital** K_t^D : requires no monitoring cost.

Payoffs at $t + 1$:

- ▶ **Safe capital**: produces $\frac{1}{\beta} K_t^S$ in both states h and l .
- ▶ **Risky capital**: produces $\frac{A_h}{\beta} K_t^D$ in state h and 0 in state l .
- ▶ **Assumption**: Equal average productivity at $t + 1$.

Equal Average Productivity implies:

$$(1 - \kappa)A_h = 1.$$

Intermediaries: Institutional Constraints

Market Structure and Constraints:

- ▶ Continuum of small, price-taking intermediaries.
- ▶ Each intermediary specializes: issues either safe debt S_t or risky debt D_t .

Frictions and Limited Liability

- ▶ **Limited Liability Assumption:** Intermediaries default at $t + 1$ if payoffs cannot cover debt obligations.
- ▶ **No Equity:** Unlike Chapter 4, intermediaries cannot raise equity to buffer shocks.

Intermediaries: Safe Debt Supply (S_t)

- ▶ Intermediaries issuing S_t invest in riskless capital K_t^S subject to budget constraint:

$$(1 + \iota)P_t K_t^S = Q_t^S S_t. \quad (11.8)$$

State-Contingent Real Profits at $t + 1$:

$$\Psi_h^S = \frac{K_t^S}{\beta} - \frac{S_t}{P_h} \quad (11.9)$$

$$\Psi_l^S = \frac{K_t^S}{\beta} - \frac{S_t}{P_l}. \quad (11.10)$$

Supply Condition:

- ▶ Limited liability requires: $\min \{\Psi_h^S, \Psi_l^S\} \geq 0$.
- ▶ Therefore, using (11.8) and profits (11.10), safe debt supply is non-negative if:

$$Q_t^S \geq \beta \frac{P_t}{\min \{P_h, P_l\}} (1 + \iota). \quad (11.11)$$

Intermediaries: Risky Debt Supply (D_t)

- Intermediaries issuing D_t invest in risky capital K_t^D subject to:

$$P_t K_t^D = Q_t^D D_t. \quad (11.12)$$

State-Contingent Real Profits at $t + 1$:

$$\Psi_h^D = \frac{A_h}{\beta} K_t^D - \frac{D_t}{P_h}, \quad (11.13)$$

$$\Psi_l^D = 0. \quad (11.14)$$

Supply Condition:

- Using (11.12) and the productivity assumption $(1 - \kappa)A_h = 1$ in (11.13):

$$Q_t^D \geq \beta \frac{P_t}{P_h} (1 - \kappa). \quad (11.15)$$

Note: Safe capital K^S could also back risky securities due to $P_h \neq P_l$ variation, but we focus on $P_h = P_l$ in equilibrium.

Entry Choice: Safe vs Risky Issuance

- ▶ Intermediaries compare expected profits across the two markets.
- ▶ A key qualitative distinction:
 - ▶ safe issuance needs a *premium* to cover monitoring cost ι ,
 - ▶ risky issuance can be profitable when households value liquidity in h .
- ▶ This trade-off is the origin of pseudo-safe assets:
 - ▶ liquidity provision can be abundant in good states via risky debt,
 - ▶ but collapses in bad states due to default.

The Public Sector: Debt and Budget Constraints

- ▶ The government balance sheet consists of liabilities (B_t), interpreted as Treasury debt or central bank reserves. These are inherently risk-free as they define the unit of account (see Chapter 1).

Budget Constraint at t :

$$Q_t^B B_t = B_{t-1} - T_t. \quad (11.16)$$

Budget Constraint at $t + 1$:

- ▶ Taxes must repay the nominal debt in both states:

$$T_h = T_l = B_t. \quad (11.17)$$

Policy Rules: Monetary and Fiscal Stance

- ▶ We assume, as in Chapter 4, the government to follow these policies:

Monetary Policy

- ▶ The central bank sets the interest rate on reserves at $1 + i_t^X = 1/\beta$.
- ▶ **Implication:** The bond price is anchored at $Q_t^B = \beta$.

Fiscal Policy (Constant Real Taxes)

- ▶ The treasury maintains a constant real tax burden $b > 0$:

$$\frac{T_t}{P_t} = (1 - \beta)b, \quad (11.18)$$

$$\frac{T_h}{P_h} = \frac{T_l}{P_l} = b, \quad (11.19)$$

for some $b > 0$.

Price Level Determination

The interaction of fiscal and monetary policies determines the path of prices.

Constant Price Level at $t + 1$

- ▶ Combining (11.17) and (11.19), the price level is equal across contingencies:

$$P_h = P_l = P^*, \quad \text{where} \quad P^* = \frac{B_t}{b}.$$

Determining the Price Level at t

- ▶ Substituting the tax policy (11.18) into the budget constraint (11.16) yields:

$$\frac{B_{t-1} - \beta B_t}{P_t} = (1 - \beta)b. \quad (11.20)$$

Note on Uniqueness

- ▶ (11.20) provides one restriction on two unknowns (B_t and P_t). Uniqueness requires an additional restriction coming from equilibrium in the liquidity market (through q_h, q_l and asset demands).

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity**
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Public Satiation of Liquidity

We first analyze a baseline where the government supply of liquidity is ample enough to satisfy all domestic liquidity needs, rendering private supply irrelevant.

The Satiation Condition

- ▶ Satiation occurs if the real tax policy b is sufficiently high ($b \geq \bar{q}$).
- ▶ **Implication:** The marginal utility of liquidity is zero in all states:

$$V_q(\cdot) = 0 \implies \varrho_h = \varrho_l = 0.$$

Asset Pricing and Price Stability

- ▶ With $\varrho = 0$ and the interest-rate policy $Q_t^B = \beta$, the FOC (11.5) simplifies to:

$$\beta = \beta \left[(1 - \kappa) \frac{P_t}{P^*} + \kappa \frac{P_t}{P^*} \right] \implies P_t = P^*.$$

- ▶ The price level is perfectly stable over time under full government satiation.

Stationary Equilibrium and "Crowding Out"

Equilibrium Debt and Price Level

- ▶ Using $P_t = P^* = B_t/b$ in the budget restriction (11.20):

$$\frac{B_{t-1} - \beta B_t}{B_t/b} = (1 - \beta)b \implies B_t = B_{t-1}.$$

- ▶ Consequently, the price level is determined by the initial debt stock: $P^* = B_{t-1}/b$.

Crowding Out of Private Safe Assets

- ▶ If $\iota > 0$, intermediaries will **not** produce safe assets S_t .
- ▶ **Logic:** At satiation, there is no liquidity premium ($\varrho = 0$). Intermediaries cannot sell S_t at a price high enough to cover the monitoring cost ι .
- ▶ *Result:* Only government liquidity and potentially defaulted private securities exist.

Discussion: Public Liquidity and Fiscal Limits

Interpretation

- ▶ In this benchmark equilibrium, the government supplies enough **public liquidity** to fully satiate the economy's liquidity needs, so that liquidity premia vanish.
- ▶ Private creation of safe liquidity becomes **irrelevant** (and, with monitoring costs, is crowded out).

Critical Assumptions

- ▶ The feasibility of full liquidity satiation rests on two key requirements:
 - 1 **Benevolence**: the government acts to maximize household welfare.
 - 2 **Fiscal capacity**: there are no effective limits on raising (state-contingent) taxes to back public liabilities.

Preview: Relaxing the "No Tax Limit" Assumption

If fiscal capacity is limited ($b < \bar{q}$), a shortage arises. However, other interventions can mitigate the resulting liquidity crises:

- ▶ **Alternative Tools with $b < \bar{q}$:**
 - ▶ **Asset Purchases:** Central bank holds risky assets in its portfolio.
 - ▶ **Guarantees:** Deposit insurance or public backstops.
 - ▶ **Regulation:** Direct oversight of intermediary portfolio choices.
- ▶ **Key Idea:** Risky assets provide income in good states (allowing lower taxes), but default in bad states—requiring fiscal backing exactly when crises occur.

Next: We restrict public supply ($b < \bar{q}$) to study the private creation of liquidity.

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets**
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Frictionless Private Supply ($\iota = 0$)

In this framework, we assume no monitoring costs for investing in risk-free projects ($\iota = 0$).

- ▶ Free entry and competition reduce all intermediary profits to zero.
- ▶ This implies the supply of safe (11.11) and risky debt (11.15) is non-negative at prices:

$$Q_t^S = \beta \frac{P_t}{P^*}, \quad Q_t^D = \beta \frac{P_t}{P^*} (1 - \kappa). \quad (11.21)$$

- ▶ These equations hold with equality in equilibrium due to free competition.

Why an Equilibrium with Only Risky Debt Cannot Exist

Why must private safe debt be supplied?

- ▶ Suppose (by contradiction) intermediaries issue only risky debt D .
- ▶ In the bad state, risky debt defaults $\Rightarrow$ liquidity equals public liquidity:
 $\varrho_l = V_q(b) > 0$.
- ▶ Households would attach a high value to safety, willing to pay:

$$Q_t^S = \beta (1 + \kappa \varrho_l) \frac{P_t}{P^*} > \beta \frac{P_t}{P^*}.$$

- ▶ With $\iota = 0$, the cost of supplying S_t is only $\beta P_t / P^*$. Thus, safe issuance would yield strictly positive profits, i.e. $\kappa \varrho_l \frac{P_t}{P^*} > 0$.

Contradiction: Competition drives entry until safe debt is supplied up to the point where the premium ϱ_l is eliminated.

Liquidity Satiation and the First Best

Because the private sector can supply "safe assets", the economy always achieves the first best ($\bar{q}$), regardless of government fiscal capacity.

The Resulting Liquidity Premia

- ▶ Combining supply (11.21) with household demand (11.6), we obtain:

$$q_h = q_l = V_q \left(\frac{B_t + S_t}{P^*} \right) = 0.$$

- ▶ Even if public liquidity is scarce ($b < \bar{q}$), safe private debt S_t expands to fill the gap.
- ▶ **No Liquidity Crunch:** Total safe assets remain constant across states h and l .

Price Level and Frictionless Equivalence

Price Level Determination

- ▶ Given $Q_t^B = \beta$ and $Q_t^S = \beta P_t / P^*$, in equilibrium $P_t = P^*$.
- ▶ From the anchor of the price level (11.20), $B_t = B_{t-1}$ and the price level is determined by: $P^* = B_{t-1} / b$.

Summary: Equivalence Result

- ▶ Frictionless private money creation does **not** impair price-level control.
- ▶ Private liquidity replicates the allocation achieved by full government provision.
- ▶ The composition of the "Safe Asset" pool is irrelevant for welfare.

Next: We introduce monitoring costs ($\iota > 0$), where private safe production becomes costly and "pseudo-safe" assets emerge.

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets**
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

Monitoring Costs and the Safe Asset Shortage

We now analyze the case where intermediaries face a positive monitoring cost $\iota > 0$ when investing in risk-free projects.

Possible Market Scenarios

- ▶ Three scenarios are theoretically possible:
 - ① **Safe-only:** All intermediaries issue safe debt (S_t).
 - ② **Risky-only:** All intermediaries issue risky debt (D_t).
 - ③ **Mixed:** Both types of securities co-exist.
- ▶ We begin by proving, by contradiction, why a "safe-only" scenario cannot be an equilibrium.

Infeasibility of a “Safe-only” Equilibrium (I)

Suppose, by contradiction, that all intermediaries issue only safe debt (S).

- ▶ To break even, safe debt must command a liquidity premium to cover $\iota > 0$.
- ▶ Equating demand (11.6) and supply (11.11) implies:

$$\varrho_h = \varrho_l = \iota > 0. \quad (11.22)$$

- ▶ **Logic:** If $\varrho = 0$, intermediaries would make negative profits because the market price would not compensate for the monitoring cost ι .

Infeasibility of a “Safe-only” Equilibrium (II)

Liquidity Below Satiation

- ▶ A positive premium ($\varrho > 0$) indicates that liquidity is below the first-best level: $q < \bar{q}$.

Symmetry Across States

- ▶ Since only safe securities (which provide equal services in both states) are issued, real liquidity is equalized:

$$q_h = q_l < \bar{q}. \quad (11.23)$$

- ▶ Consequently, state h remains underserved, resulting in a positive liquidity premium: $\varrho_h > 0$.

Infeasibility of a “Safe-only” Equilibrium (III)

Entry of Risky Debt

- ▶ Because $\varrho_h > 0$, a deviating intermediary can issue **risky debt** (D_t):
 - ▶ It requires **no monitoring costs** ($\iota = 0$).
 - ▶ It satisfies the liquidity demand in state h .

The Contradiction

- ▶ The intermediary earns strictly positive profits in the high state:

$$\Psi_h^D = \varrho_h \frac{D_t}{P^*} > 0. \quad (11.24)$$

- ▶ **Conclusion:** The "safe-only" scenario is not an equilibrium. Risky debt will always be supplied to capture these rents, i.e. $D > 0$.

Equilibrium Satiation in State h

The "profitable deviation" logic extends to any scenario where $\varrho_h > 0$. Intermediaries will continue to supply risky debt until all potential rents are exhausted.

The Satiation Result

- Competition in the supply of risky debt ensures that the liquidity premium in the good state vanishes:

$$\varrho_h = 0. \quad (11.25)$$

- **Efficiency:** Liquidity is at the first-best level in state h :

$$q_h = \bar{q}. \quad (11.26)$$

The Supply of Private Safe Assets

We have established that risky debt will be supplied. Will private safe debt be supplied as well?

The Profitability Condition

- ▶ The answer is affirmative only if intermediaries can issue them at a premium to offset the monitoring cost.
- ▶ This depends in turn on the amount of **public liquidity**:
 - ▶ **Large public supply**: Low liquidity premium on safe debt $\Rightarrow$ issuing safe debt is not profitable.
 - ▶ **Low public supply**: Creates a profitable opportunity for intermediaries to issue safe debt.

The Crowding-Out Condition (I)

Intermediaries will only supply safe debt if the price investors are willing to pay covers the monitoring cost.

Supply and Demand Prices

- To cover investment and monitoring costs, supply (11.5) requires:

$$Q_t^S = \beta (1 + \iota) \frac{P_t}{P^*}. \quad (11.27)$$

- However, from (11.6), investors will **not** hold safe private debt if the price is higher than:

$$\tilde{Q}_t^S = \beta (1 + \kappa \varrho_l) \frac{P_t}{P^*}. \quad (11.28)$$

- Therefore, a low liquidity premium in state l makes it harder to supply safe securities.

The Crowding-Out Condition (II)

The Crowding-Out Result

- ▶ Safe private debt will **not** be supplied if the required supply price exceeds the investor's limit ($Q_t^S > \tilde{Q}_t^S$).
- ▶ This occurs when the public liquidity (b) is high enough that:

$$V_q(b) < \frac{l}{\kappa}. \quad (11.29)$$

- ▶ **Conclusion:** Sufficiently high public debt reduces the liquidity premium to a point where private production of safety is no longer profitable.

Equilibrium Liquidity Premium in the Bad State

The previous feasibility logic implies:

$$q_l = \min \left\{ \frac{l}{\kappa}, V_q(b) \right\}. \quad (11.30)$$

Hence real liquidity in the bad state:

$$q_l = \max \left\{ V_q^{-1} \left(\frac{l}{\kappa} \right), b \right\} < \bar{q}.$$

- ▶ If public liquidity is relatively abundant (so $V_q(b)$ is small), safe private debt is *crowded out*.
- ▶ If public liquidity is scarce (large $V_q(b)$), some safe private debt is issued to cover monitoring costs.

Equilibrium Premium and Asset Supply

- **Safe Private Debt (S):** Supplied only if public debt b is below the private production threshold:

$$S = P^* \max \left(V_q^{-1} \left(\frac{l}{\kappa} \right) - b, 0 \right). \quad (11.31)$$

- **Risky Private Debt (D):** Always positive ($D > 0$) to ensure satiation in state h , at prices $Q^D = (1 - \kappa) P_t / P^*$:

$$D \geq P^*(\bar{q} - b) - S. \quad (11.32)$$

Private Asset Quantities

The presence of a liquidity premium influences the price level at time t relative to $t + 1$.

- ▶ Since $Q_t^B = \beta (1 + \kappa \varrho_l) P_t / P^*$ and $Q_t^B = \beta$, the price level at time t is lower than the future price level P^* :

$$P_t = \frac{P^*}{(1 + \kappa \varrho_l)} < P^*.$$

- ▶ Using $P^* = B_t / b$ and the budget restriction (11.20), we find that $B_t > B_{t-1}$:

$$\frac{B_t}{B_{t-1}} = \frac{1 + \kappa \varrho_l}{1 + \beta \kappa \varrho_l}.$$

Solving for P_t and P^*

$$P_t = \frac{1}{1 + \beta \kappa \varrho_l} \frac{B_{t-1}}{b}, \quad P^* = \frac{1 + \kappa \varrho_l}{1 + \beta \kappa \varrho_l} \frac{B_{t-1}}{b}.$$

Seigniorage and the Intertemporal Budget Constraint

The government issues more debt because the liquidity premium allows it to borrow at a cost lower than the market risk-free rate.

Intertemporal Resource Constraint (see Chapter 4)

$$\frac{B_{t-1}}{P_t} = \frac{T_t}{P_t} + (Q_t^B - Q_t^f) \frac{B_t}{P_t} + \beta \left(\kappa \frac{T_h}{P_h} + (1 - \kappa) \frac{T_l}{P_l} \right). \quad (11.33)$$

Liquidity Revenues

- ▶ Let $Q_t^f = \beta P_t / P^*$ be the price of a risk-free but **non-liquid** security.
- ▶ Since $Q_t^f < Q_t^B$, the term $(Q_t^B - Q_t^f)$ represents extra “**revenues**” to back the price level.
- ▶ **Result:** With a fixed tax policy, these extra revenues allow for a lower price level at time t .

Monetary Policy and the Price Level

The dependence of P_t on ϱ_l implies that shifts in monitoring costs (ι) can influence the price level, posing a challenge for monetary policy.

Price Level Control

- ▶ This sensitivity is a feature of the specific tax policy, not a general result.
- ▶ Control is restored by adjusting the tax policy at $t + 1$:

$$\frac{T_h}{P_h} = \frac{T_l}{P_l} = b - (Q_t^B - Q_t^f) \frac{B_t}{P_t}.$$

- ▶ **Result:** Inserting this into the IBC (11.33) reveals that P_t returns to the same level P^* as in the frictionless allocation.

Interest Rate Policy for Price Stability

To ensure prices remain at P^* at time $t + 1$, the government must also adjust its interest rate policy.

The Stabilization Rate

- ▶ The central bank sets the interest rate on reserves to:

$$1 + i_t^X = \beta^{-1} (1 + \kappa_{Ql})^{-1}.$$

- ▶ This sets the price of government bonds to $Q_t^B = \beta (1 + \kappa_{Ql})$.

Constant Price Level

- ▶ Combined with the household FOC (11.5), this implies $P_t/P_{t+1} = 1$, ensuring a constant price level.
- ▶ In this equilibrium, $B_t < B_{t-1}$, consistent with the lower taxes levied at $t + 1$ to offset liquidity revenues.

Monetary Policy and Private Liquidity

- ▶ Inflation control remains feasible even when liquidity is not fully satiated (consistent with Chapter 3).
- ▶ The key difference is that the central bank must now account for **private** safe-asset supply when setting the policy rate.

Interest-rate adjustment for price stability

- ▶ To keep the price level (or inflation) on target, the interest rate on reserves must be set **inversely** to the liquidity premium in the low state, ϱ_l .
- ▶ **Implementation logic:**
 - ▶ Higher monitoring costs ($\iota \uparrow$) make safe private issuance more expensive, increasing the scarcity premium ($\varrho_l \uparrow$).
 - ▶ To offset the higher liquidity premium and maintain price stability, the central bank must **lower** the policy rate.
- ▶ A practical constraint is the **zero lower bound**, which can limit how far the policy rate can fall when ϱ_l rises sharply.

The Zero Lower Bound (ZLB)

Frictions in private safe-asset production create specific constraints for the policy rate.

Monitoring Costs and the ZLB

- ▶ Sudden variations in the cost of monitoring may require a rate cut that is constrained by the **zero-lower bound**.

Public Debt as a Buffer

- ▶ The premium ϱ_l is bounded above by the supply of government liquidity.
- ▶ **Key Result:** A relatively higher supply of public debt limits the extent to which the premium can rise, thereby limiting the necessary downward correction of the policy rate.

Conclusion: The Endogenous Liquidity Crunch

- ▶ The central result is an equilibrium with **state-contingent liquidity**: liquidity is abundant in good states but scarce in bad states.

Interpretation

- ▶ **Good state**: abundant issuance of *pseudo-safe* claims makes liquidity appear fully satiated.
- ▶ **Bad state**: pseudo-safe assets default and cease to provide liquidity services $\Rightarrow$ an endogenous **shortage of safe assets**.

Real economic costs

- ▶ The liquidity crunch reduces the real value of transaction and collateral services.
- ▶ In richer models where liquidity affects production, the collapse in liquidity can translate into lower aggregate activity (Benigno and Nisticò 2017).

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention**
- 8 General Discussion
- 9 Main Takeaways

Why Intervene?

The Failure of *Laissez-Faire* ($\iota > 0$)

- ▶ Liquidity is sufficient in state h (first-best).
- ▶ Liquidity collapses in state l due to defaults on risky debt.

Government Objectives

- ▶ **Crisis Prevention:** Ensure $q_l = \bar{q}$ to eliminate the liquidity crunch.
- ▶ **Fiscal Efficiency:** Achieve satiation while economizing on taxes under limited fiscal capacity.

Four Policy Approaches

- 1 **Large Public Liquidity:** Satiation via high taxes.
- 2 **Asset Purchases:** Central Bank balance sheet management.
- 3 **Guarantees:** Actuarially fair deposit insurance.
- 4 **Regulation:** Portfolio restrictions forcing safe investment.

Policy Equivalence and Efficiency

The first best can be implemented without high average taxes by exploiting the backing provided by private intermediaries in good times.

Wallace (1981) Equivalence

- ▶ **Asset Purchases** and **Deposit Insurance** are functionally equivalent.
- ▶ Both policies require government backing *only* in bad times (l).
- ▶ **Fiscal Identity**: The tax burden across all contingencies and the consolidated balance sheets of liquidity providers are identical under both regimes.

The Regulatory Trade-off

- ▶ Regulation forcing safe investment **reduces welfare**.
- ▶ **Rationale**: It prevents the economy from economizing on monitoring costs (l) by restricting the issuance of risky debt in favorable states.

Government policy with a limit on taxes

If the government faces a limit on its average real tax capacity, it can no longer satiate liquidity purely through taxation.

The Tax Limit

$$(1 - \kappa) \frac{T_h}{P_h} + \kappa \frac{T_l}{P_l} \leq b < \bar{q}. \quad (11.34)$$

The Solution: Central Bank Asset Purchases

- ▶ The first best ($q_h = q_l = \bar{q}$) is still achievable by purchasing private risky debt D_t^C .
- ▶ **Friedman (1960) Proposal:** Backing interest-bearing reserves (B) with a portfolio of private risky assets.
- ▶ **The State-Contingent Mechanism:**
 - ▶ **High state (h):** Asset payoffs allow for *lower* taxes.
 - ▶ **Low state (l):** Assets default; backing shifts to the fiscal authority (taxes).

Asset Purchases: Central Bank Balance Sheet at t

The central bank purchases a quantity D_t^C of risky securities and finances these purchases by increasing outstanding debt from B_t to $\bar{B}$.

The Budget Constraint at time t

- ▶ The nominal budget constraint reflects the acquisition of risky assets:

$$Q_t^B \bar{B} - Q_t^D D_t^C = B_{t-1} - T_t. \quad (11.35)$$

- ▶ Substituting the tax policy $T_t/P_t = (1 - \beta)b$ and $B_t = \bar{B}$, the real budget constraint is:

$$\frac{Q_t^B \bar{B} - Q_t^D D_t^C}{P_t} = \frac{B_{t-1}}{P_t} - (1 - \beta)b. \quad (11.36)$$

Asset Purchases: Solvency at $t + 1$

At time $t + 1$, the government repays the debt $\bar{B}$ using the proceeds earned by holding assets D_t^C (if not in default) and taxes.

State-Contingent Repayment

- **High State (h):** The risky debt held by the central bank pays a return, allowing for a reduction in taxes:

$$\bar{B} = T_h + D_t^C. \quad (11.37)$$

- **Low State (l):** Private risky securities are defaulted on; debt requires full backing through taxes:

$$\bar{B} = T_l. \quad (11.38)$$

Implementing the First Best: Fiscal Constraints

To achieve the first best ($q_h = q_l = \bar{q}$), the government must back a real debt level of $\bar{q}$ in both states.

The Maximum Fiscal Effort in State l

- In the low state, risky assets default. To pay back debt, the government raises taxes to the maximum level:

$$\frac{T_l}{P_l} = \bar{q} \implies P_l = \frac{\bar{B}}{\bar{q}}. \quad (11.39)$$

The Residual Capacity in State h

- Given the average tax limit $(1 - \kappa)\frac{T_h}{P_h} + \kappa\frac{T_l}{P_l} \leq b$, and substituting $T_l/P_l = \bar{q}$, the taxes in the high state are constrained to:

$$\frac{T_h}{P_h} = \frac{b - \kappa\bar{q}}{1 - \kappa} < \bar{q}. \quad (11.40)$$

Implementing the First Best: Required Asset Purchases

Since T_h/P_h is insufficient to back $\bar{q}$ alone, the government uses asset purchases to bridge the gap.

Asset Quantity (D^C)

- ▶ Substituting the tax result into the state- h budget (11.37):

$$\frac{\bar{B}}{P_h} = \frac{b - \kappa \bar{q}}{1 - \kappa} + \frac{D^C}{P_h}. \quad (11.41)$$

The Efficiency Condition

- ▶ Implementing the first best requires prices to be equalized across states ($P_h = P_l$). Solving for D^C yields:

$$D^C = \frac{\bar{q} - b}{1 - \kappa} \frac{\bar{B}}{\bar{q}}. \quad (11.42)$$

- ▶ **Intuition:** The central bank "manufactures" safety by collecting risky returns in state h to offset the fiscal shortage.

Equilibrium Prices and Stability

Under the intervention regime, the economy achieves the first-best allocation, removing the frictions that drove the *laissez-faire* price level.

Vanishing Premia

- ▶ Because $q_h = q_l = \bar{q}$, the marginal utility of liquidity is zero:

$$\varrho_h = \varrho_l = 0. \quad (11.43)$$

Price Level Determination

- ▶ With $Q_t^B = \beta$ and zero premia, the household FOC (11.5) implies:

$$P_t = P_h = P_l = P^{**}. \quad (11.44)$$

- ▶ **Result:** Asset purchases restore a constant price level across all states and time periods.

Balance-Sheet Expansion

Solving for $\bar{B}$

- ▶ Substitute $Q^D = \beta(1 - \kappa)$ and the derived D^C into the real budget (11.36):

$$\frac{\beta\bar{B} - \beta(1 - \kappa)D^C}{P^{**}} = \frac{B_{t-1}}{P^{**}} - (1 - \beta)b \implies \bar{B} = \frac{\bar{q}}{b}B_{t-1}. \quad (11.45)$$

Key Conclusions

- ▶ **Debt Expansion:** Nominal government liabilities increase to finance the purchases of risky assets ($\bar{B} > B_{t-1}$ since $\bar{q} > b$).
- ▶ **Price Level:** The equilibrium price level P^{**} is higher than in the shortage benchmark because the nominal scale of liabilities is larger.
- ▶ **Fiscal Backstop:** Crises are avoided as long as the government can raise taxes in the bad state l up to $\bar{q}$.

Deposit Insurance and Wallace Equivalence

The first best is also implementable via an **actuarially fair deposit insurance** program that guarantees intermediary liabilities.

The Mechanism: State-Contingent Transfers

- ▶ **State h :** The government charges intermediaries a fee; there is no subsidy on average.
- ▶ **State l :** Private payoffs are zero; the government transfers resources to intermediaries to ensure debt is repaid in full.
- ▶ **Result:** Intermediary debt is "hardened" into a safe asset, providing liquidity services despite risky underlying investments.

The Wallace (1981) Logic

- ▶ **Equivalence:** Under both Asset Purchases and Deposit Insurance, the **consolidated balance sheet** of "liquidity suppliers" (Gov + Intermediaries) is identical.
- ▶ Therefore, the same consumption allocation, prices, and taxes are implementable.

Challenging the “Bills Only” Doctrine

This equivalence has significant implications for how central banks manage their balance sheets, even in normal times.

Asset Composition and the Tax Burden

- ▶ **Conventional View:** Central banks should hold only safe Treasury bills. Under this approach, no reduction in the tax burden is possible.
- ▶ **Unconventional View:** By holding (or guaranteeing) private risky securities, the central bank fulfills liquidity needs while **economizing on taxes** in good times.

The Fiscal Bottom Line

- ▶ A crisis is **unavoidable** if the government cannot raise taxes in state l .
- ▶ **The Friction:** Private safety is costly to manufacture ($\iota > 0$); public liquidity can crowd out this private safe debt, making government backing even more critical.

Theory and Practice: Fundamental Shocks

The role of deposit insurance here extends beyond the classic Diamond and Dybvig (1983) framework.

Fundamental Shocks vs. Panics

- ▶ **Standard View:** Insurance prevents "bad equilibria" (panics/runs).
- ▶ **This Framework:** Insurance mitigates **fundamental shocks** (the state l default) that would otherwise trigger a systemic liquidity collapse.

Historical Evidence (2008 Crisis)

- ▶ **United States:** FDIC limits increased (\$100k to \$250k) and the *Temporary Liquidity Guarantee Program* was created to "harden" liabilities.
- ▶ **Germany:** Provided **unlimited** coverage to ensure the private pool of safe assets did not evaporate.

Regulation of Intermediaries' Investments

In the aftermath of the 2008 crisis, further regulation of financial intermediaries emerged.

Modelling

- ▶ Intermediaries are forced to invest **only** in riskless projects (K^S).
- ▶ **Result:** Private liabilities become inherently safe; the need for government guarantees or deposit insurance is eliminated.

Fiscal Capacity vs. Private Backing

- ▶ **Public Backing:** Asset purchases and guarantees rely on the government's fiscal capacity to absorb shocks in state l .
- ▶ **Regulation:** Changes the **private backing** of securities directly through portfolio restrictions, requiring zero fiscal capacity from the sovereign.

The Welfare Cost of Mandated Safety

While regulation eliminates liquidity crises, it is fundamentally welfare-inferior to the *laissez-faire* equilibrium.

The Resource Waste

- ▶ *Laissez-faire*: Intermediaries choose risky technologies to **economize on monitoring costs** (ι). This results in "pseudo-safe" debt that defaults in state l but is cheap to produce.
- ▶ **Regulation**: Forces the payment of monitoring costs on *all* issued liquidity.

Core Conclusion

- ▶ Regulation trades off a liquidity crisis for a **waste of real resources**.
- ▶ Benigno and Robatto (2019) prove that *laissez-faire* is constrained-efficient; thus, regulation reduces welfare by over-investing in costly safety.

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion**
- 9 Main Takeaways

Control of the Price Level

- ▶ A central concern is whether private money creation and the existence of “pseudo-safe” assets generate **price-level indeterminacy**.

Robustness of nominal control

- ▶ **Result:** an expansion of money-like liabilities through private intermediaries does **not** necessarily imply indeterminacy, provided the price level is anchored according to the principles in Chapters 1–2.
- ▶ **Frictionless benchmark:** as in Chapter 4, competitive private liquidity provision can achieve full satiation without undermining price stability.
- ▶ **With frictions:** the central bank can still maintain the desired price level by appropriately adjusting the interest rate on reserves (taking private liquidity conditions into account).

The Limits of Monetary Policy

- ▶ Interest-rate policy can stabilize *prices*, but it has structural limits in a liquidity crisis.

Nominal stability vs. real contraction

- ▶ **Real liquidity failure:** interest-rate policy *alone* cannot prevent a contraction in **real liquidity** when pseudo-safe debt defaults and stops providing liquidity services.
- ▶ **Zero lower bound:** the rate cuts needed to offset a sharp rise in liquidity premia may be constrained by $i_t \geq 0$.
- ▶ **Public buffer:** one way to mitigate the constraint is to expand **public liquidity** in adverse states, crowding out costly private safe-asset production and sustaining safe liquidity provision.

Conclusion

- ▶ Balance-sheet policies and fiscal backstops matter for stabilization because they address the **quantity** (backing) of safe assets, not only their **price** (interest rate).

Supply of Liquidity: The Classic Benchmarks

In a frictionless world ($\iota = 0$) with no tax limits, both private and public systems can achieve the first best (consistent with Chapter 4).

Hayek (1976): Free Banking

- ▶ Competition leads the private sector to supply sufficient "safe assets."
- ▶ **Alignment of Interests:** The liquidity premium reduces intermediaries' financing costs; thus, their profit motive aligns with society's need for liquidity.
- ▶ Free entry ensures the supply of safe securities fully satiates demand ($q = \bar{q}$).

Friedman (1960): Narrow Banking

- ▶ The government holds a monopoly on liquidity via a **100% reserve requirement**.
- ▶ Intermediaries hold safe government debt (B) instead of illiquid capital; the government backs this debt through taxes.
- ▶ If the benevolent government faces no limit on taxes, it achieves the first best by supplying $b \geq \bar{q}$.

The Failure of Pure Systems under Frictions

With monitoring costs ($\iota > 0$) and tax limits ($b < \bar{q}$), neither "pure" system can satiate the demand for liquidity alone.

The Private Market Failure

- ▶ Frictions incentivize the creation of **risky debt** to avoid monitoring costs.
- ▶ This generates an equilibrium that fails the first best and remains prone to liquidity crises.

The Public Fiscal Limit

- ▶ Friedman's narrow banking fails if there are limits on taxes in bad states (l).
- ▶ **Conclusion:** If the government cannot back its debt in state l , it cannot avert the drop in real liquidity. Optimal policy represents a **hybrid mix** of the two views.

Policy Response: Nominal Targets in Crises

When a crisis originates from the poor quality of private assets, policy must act to maintain the economy's macroeconomic trajectory.

The Objective: Nominal Stability

- ▶ Monetary policy should keep the economy on its pre-crisis trajectory with respect to nominal targets, such as **nominal spending**.

Implementation Mechanisms

- ▶ **Interest Rates:** Lowering the policy rate to push down all money market rates and reduce liquidity premia.
- ▶ **Balance Sheet:** Maintaining liquidity by supplementing deteriorated private assets with high-quality central-bank (government) securities.

Caution on Regulation and "Real Bills"

Modern regulations (e.g., **Basel III**) often mirror the "**Real Bills**" doctrine, requiring intermediaries to hold essentially risk-free assets (Sargent 2011).

The Welfare Trade-off

- ▶ **The Goal:** Prevent crises by ensuring real liquidity is safe in both good and bad states.
- ▶ **The Cost:** Forcing investment in safer but **lower productive assets** can be welfare-negative relative to *laissez-faire*.

Economic Logic

- ▶ Regulation trades a liquidity crisis for a persistent waste of resources (monitoring costs ι).
- ▶ Efficiency may require allowing "pseudo-safe" debt in good times, provided a credible government backstop exists for bad times.

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways**

Main Takeaways: Safe Assets and Liquidity Crises

- ▶ **Safe assets** deliver liquidity services; scarcity implies a **liquidity premium** and low safe rates.
- ▶ In the **frictionless benchmark** ($\iota = 0$), private intermediaries supply enough safe debt to **sate** liquidity in all states, and the central bank can still **pin down the price level**.
- ▶ With **monitoring costs** ($\iota > 0$), truly safe private debt is limited and the economy relies on **pseudo-safe** claims: liquidity is abundant in good states but collapses in bad states when risky debt defaults.
- ▶ **Monetary policy** can stabilize the **price level** by adjusting the policy rate (taking private liquidity conditions into account), but it cannot by itself prevent the **drop in real liquidity** in the crisis state (and the ZLB may bind).
- ▶ **Policy responses**: asset purchases and actuarially fair guarantees can eliminate liquidity crises (and are equivalent under consolidated balance sheets), while regulation can remove crises at a **resource cost** (welfare loss).

- 1 Introduction
- 2 Outline of the results
- 3 Model
- 4 Equilibrium with the Government Satiating Liquidity
- 5 Equilibrium with No Frictions in the Supply of Private Safe Assets
- 6 Equilibrium with Frictions in the Supply of Private Safe Assets
- 7 Government Intervention
- 8 General Discussion
- 9 Main Takeaways

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